



Advanced MCU Digital Watch & Environmental Station

An Industrial-Grade, Non-Blocking Multi-Function Station Featuring Real-Time I2C RTC, Dual Telemetry ADC, Priority Dual Alarms, and Anti-Flicker State Machine



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🔗 SYSTEM ARCHITECTURE & PURPOSE

Project Overview & Core Engineering Vision

Transforming a conventional microcontroller clock into a deterministic, multi-tasking environmental station

🎯 Project Objectives

- ✅ **Ultra-Accurate Timekeeping:** Hardware RTC DS1307 battery-backed clock with dual 12H/24H formats.
- ✅ **Multi-Mode Platform:** 6 interactive screens (Clock, Telemetry, Alarms, Countdown, Stopwatch, Pomodoro).
- ✅ **Zero-Flicker Display:** Smart differential rendering pipeline over 4x20 Character LCD.

⚠️ Real-Time Challenges

- ⚠️ **Deterministic Concurrency:** Running countdowns, alarms, and stopwatches concurrently without blocking `_delay_ms()`.
- ⚠️ **ADC Noise Filtering:** Accurate LM35 temperature and analog humidity sampling on 10-bit ADC.
- ⚠️ **Keypad Debouncing:** Finite State Machine key scanning with rapid responsiveness.

🏆 Key Engineering Highlights

- ★ **Layered Architecture:** Strict separation of MCAL, HAL, and Application layers.
- ★ **Timer0 Master Scheduler:** 31.25 Hz hardware interrupt tick managing asynchronous state engines.
- ★ **Global Settings Engine:** Dynamic configuration overlay accessible instantly via the `[*]` key.



SOFTWARE ENGINEERING

Modular Layered Software Architecture

Adhering to Automotive & Industrial Embedded Standards: Complete Hardware Abstraction

APPLICATION LAYER (APP)

State Machines, Screen Handlers, Background Services & UI Engine

App_Clock

App_Telemetry

App_Alarm

App_Timer

App_Stopwatch

App_Pomodoro

HARDWARE ABSTRACTION LAYER (HAL)

Device-Specific Drivers, Protocol Wrappers & Filtering

LCD 4×20 (HD44780)

Keypad 4×4

RTC DS1307

Active Buzzer

MICROCONTROLLER ABSTRACTION (MCAL)

Direct Register Access, Peripherals & Low-Level ISRs

DIO (GPIO)

ADC (10-bit)

I2C / TWI

TIMER0 (Tick Engine)

TARGET HARDWARE PLATFORM

Physical Silicon, Quartz Oscillators & Sensor Array

ATmega32 @ 8MHz

DS1307 + 32.768kHz

LM35 Temp



Design Principles



Portability & Reusability: Application algorithms are 100% target-agnostic. MCAL can be ported to STM32/ESP32 with zero app changes.



Information Hiding: Application never touches hardware registers directly (PORTA, TWCR, ADCSRA).



Static Memory Allocation: Zero heap memory allocation (malloc), eliminating memory fragmentation.



 HARDWARE & SCHEMATICS

Hardware Interfacing & MCU Pin Allocation

ATmega32 40-Pin DIP Pinout Mapping & Bus Topology



Port A & Port B Mapping

PIN	PERIPHERAL	FUNCTION
PA0 (ADC0)	LM35 Sensor	Analog Temp In
PA1 (ADC1)	Humidity Sensor	Analog RH% In
PA2 - PA7	LCD 4×20	Data (D2-D7)
PB0 - PB3	LCD Control	RS, RW, EN, D0



Port C & I2C Bus Topology

PIN	PERIPHERAL	FUNCTION
PC0 (SCL)	DS1307 RTC	I2C Serial Clock
PC1 (SDA)	DS1307 RTC	I2C Serial Data
PC2 (DI0)	Active Buzzer	Alarm & Alerts
PC3 - PC7	Status LEDs	Telemetry Indicators



Port D (4x4 Keypad)

PIN	KEYPAD LINE	FUNCTION
PD0 - PD3	Rows R1 - R4	Input (Pull-up)
PD4 - PD7	Cols C1 - C4	Output (Drive Low)
Protocol	Matrix Scan	15ms Debounce



🕒 SCREEN 1 • MAIN TIMEKEEPING

[Screen 1] Real-Time Clock & Dynamic Calendar

Hardware I2C Communication with DS1307 RTC, BCD Translation, & Dual Format Render



Live 4x20 Character Display

```
=====
TIME: 10:45:32 PM
DATE: 28/08/2026
DAY : FRIDAY [24H]
```

DS1307 I2C MASTER MODE @ 100 KHZ

Communicates over Two-Wire Interface (TWI) bus reading seconds, minutes, hours, day-of-week, date, month, and year from register addresses 0x00 through 0x06.



Technical Implementation

- ↔ **BCD to Binary Math:** Real-time conversion: $val = ((bcd \gg 4) * 10) + (bcd \& 0x0F)$ for high-speed arithmetic.
- ⚙️ **12H / 24H Selectable Format:** Instant conversion algorithm with AM/PM indicator flag, maintained across state navigations.
- 🔋 **Non-Volatile Battery Backup:** CR2032 coin cell ensures continuous oscillation of the 32.768 kHz crystal during MCU power cycles.



[Screen 2] Climate & Environmental Telemetry

10-Bit ADC Sampling, Moving Average Filter, & Tri-Unit Thermal Conversion



Live 4x20 Character Display

```
*** CLIMATE HUB ***  
TEMP: 26.5 C (79.7F)  
HUMIDITY : 58 % RH  
STATUS   : COMFORT
```

ADC PRE-SCALER 64 (125 KHZ ADC CLOCK)

Samples ADC0 (LM35 precision analog sensor) and ADC1 (linear humidity sensor) using internal Vref with software noise rejection.



ADC Mathematics & Conversions

```
// LM35: 10 mV / 1°C, Vref = 5.0V, 10-Bit Resolution = 1024 uint16_t  
adc_val = ADC_Read(0); float temp_c = (adc_val * 500.0f) / 1024.0f;  
float temp_f = (temp_c * 9.0f / 5.0f) + 32.0f; float temp_k = temp_c +  
273.15f;
```

🔄 **Dynamic Unit Cycling:** Toggle between Celsius (°C), Fahrenheit (°F), and Kelvin (K) globally.

▼ **8-Sample Circular Buffer Filter:** Eliminates analog jitter and spurious EMI interference.



🔔 SCREEN 3 • AUDITORY & ALERT SUBSYSTEM

[Screen 3] Dual Multi-Alarm & Ringing Engine

Independent Dual Schedules, Dynamic Time Buffering, Snooze, & Non-Blocking Buzzer

🖥️ Live 4x20 Character Display

```
** ALARM CONFIG **  
ALARM 1: 07:00 [ON]  
ALARM 2: 18:30 [OFF]  
[1]SET [2]TOGGLE
```

PORTC PIN 2 ACTIVE TRANSISTOR BUZZER

Provides independent alarm registers. When current RTC time matches an enabled alarm, the system triggers the asynchronous ringing engine.

💣 Ringing & Control State Flow

- 🗂️ **Dynamic Typing Buffer:** Interactive numeric entry on Keypad for hours and minutes with cursor repositioning.
- 🔊 **Pulsed Auditory Alert:** Timer0 ISR modulates the buzzer at 2 Hz pulsed cadence, avoiding continuous annoying pitch.
- 🛏️ **Smart Snooze & Dismiss:** Pressing [+] adds +5 minutes to active alarm; pressing [/] completely disarms the active ringing state.



⌚ SCREEN 4 • PRECISION TIMING

[Screen 4] Precision Countdown Timer

Shift-Register Digit Entry, Background Timer0 Decrementer, & Expiry Buzzer



Live 4x20 Character Display

```
-- COUNTDOWN TIMER -  
  REMAIN: 00:14:59  
  STATUS: RUNNING  
[/]PAUSE [=]RESET
```

1-SECOND DETERMINISTIC DECREMENT

Allows setting hours, minutes, and seconds. Runs in background with persistent state across all screens.



Digital Entry & Execution Model

- ➔ **Shift-Register Entry Mode:** Entering digits shifts from right to left (HH:MM:SS) seamlessly mimicking commercial microwave timers.
- ▶ **Tri-State Engine:** Handles STOPPED, RUNNING, and PAUSED states cleanly via single-key keypad triggers.
- ⚠ **Completion Alert:** Emits a distinctive 5-beep sequence upon hitting 00:00:00 without stalling the main loop.



🔧 SCREEN 5 • ELAPSED TIME COUNTER

[Screen 5] High-Accuracy Digital Stopwatch

Continuous 1-Second Incremental Counter up to 99:59:59 with Background Persistence



Live 4x20 Character Display

```
== DIGITAL STOPWATCH
ELAPSED TIME:
  01:23:45
[/]START/STOP [=]RST
```

ZERO DRIFT ISR ELAPSED TRACKER

Maintains elapsed time in standard seconds accumulator variable driven by the Timer0 31.25 Hz scheduler.



Architecture Highlights

- 🛡️ **Screen-Decoupled Counting:** Start stopwatch, switch to check telemetry or clock, and return to find exact elapsed time.
- ⚡ **Atomic 32-Bit Access:** Elapsed ticks read with interrupt protection (`cli()` / `sei()`) to prevent race conditions.
- 🔄 **Instant Zero Reset:** Reset command zeros hours, minutes, seconds and flags instantaneously.



SCREEN 6 • PRODUCTIVITY ENGINE

[Screen 6] Pomodoro Productivity Engine

Automated 25m Work / 5m Short Break / 15m Long Break Cycles with Visual Progress Bar



Live 4x20 Character Display

```
POMODORO: WORK [2/4]
TIME : 18:42 REMAIN
PROG : [====>    ]
[/]START/PAUSE [=]R
```

4-STAGE STATE PROGRESSION MACHINE

Implements the classic productivity methodology in bare-metal embedded C, alerting the engineer at every interval boundary.



Cycle Progression Logic

- Work Sprint (25 Mins):** Focus timer with visual animated 10-segment custom character progress bar on LCD Line 3.
- Short Break (5 Mins):** Automatically triggered after each sprint with 2 short buzzer confirmation beeps.
- Long Break (15 Mins):** Automatically activated after completing 4 consecutive work sprints with 3 beeps.



⚙️ GLOBAL FEATURE • SYSTEM CONFIGURATION

Global Settings & Configuration Modal

Instant-Access Modal Overlay via Keypad `[\*]` Key from Any Active Screen



Live 4x20 Settings Modal

```
** SYSTEM SETTINGS *  
1.SET TIME & DATE  
2.TIME FORMAT: [12H]  
3.TEMP UNIT   : [ C ]
```

GLOBAL KEYPAD VECTOR

Pressing `[\*]` intercepts current screen context, preserves stack memory, and opens master configuration menu.



Available Customizations

- 📅 **RTC Time & Date Calibrator:** Live I2C write to DS1307 registers with numeric digit validation (0-23 hrs, 0-59 mins, valid leap years).
- 🕒 **Time Mode Selector:** Switches globally between 12-Hour (AM/PM) and 24-Hour Military format across all displays.
- 🌡️ **Temperature Metric:** Changes environmental display units between Celsius (°C), Fahrenheit (°F), and Kelvin (K).



UI/UX & HUMAN-MACHINE INTERFACE

4x4 Keypad State Machine & Navigation

Deterministic Event-Driven FSM with Anti-Flicker Differential LCD Refresh



4x4 Matrix Layout

1	2	3	+
4	5	6	-
7	8	9	/
*	0	#	=



Navigation Bindings

- [+] Key: Next Screen (1 → 2 → 3 → 4 → 5 → 6)
- [-] Key: Prev Screen (6 ← 5 ← 4 ← 3 ← 2 ← 1)
- [/] Key: OK / Start / Pause / Confirm Selection
- [=] Key: Back / Cancel / Reset Counter
- [*] Key: Open Global Settings Modal



Anti-Flicker Architecture

- No Full Screen Clears: Avoids continuous LCD_Clear() calls which cause harsh strobing.
- Dirty-Flag Buffer: Only mutated characters are written to the LCD.
- Bus Bandwidth: Reduces command overhead by >82%.



REAL-TIME SCHEDULING CORE

Concurrency Engine: Timer0 ISR & Scheduler

31.25 Hz Deterministic Hardware Master Tick Driving Parallel Tasks Without an RTOS



Timer0 Math & Tick Generation

```
/* System Frequency = 8,000,000 Hz, Prescaler = 1024 */ /* Timer Tick  
Rate = 8,000,000 / 1024 = 7812.5 Hz */ /* 8-Bit Overflow Period = 256  
Ticks */ /* Overflow Frequency = 7812.5 / 256 = 30.51 Hz (~31  
Overflows/sec) */ ISR(TIMERO_OVF_vect) { static uint8_t ovf_count = 0;  
ovf_count++; if (ovf_count ≥ 31) { ovf_count = 0; system_1sec_flag =  
1; // Master 1-Second Tick } }
```



Cooperative Multi-Task Dispatcher

- ✓ **Non-Blocking Execution:** Every application engine runs inside a short cooperative step function.
- ✓ **Volatile Thread Safety:** Flags shared between ISR and super-loop (system_1sec_flag) are declared volatile.
- ✓ **Guaranteed Response Time:** Keypad scan executes at 50 Hz, ensuring zero missed key presses.



🚩 SUMMARY & TECHNICAL ROADMAP

Conclusion, Future Roadmap & Technical Q&A

Summary of Accomplishments & Potential Next-Generation Commercial Upgrades



Accomplishments

- ✓ Modular 3-layer architecture (MCAL, HAL, APP).
- ✓ 6 concurrent feature-rich screens on 4x20 LCD.
- ✓ Robust I2C DS1307 RTC & dual ADC telemetry.
- ✓ Zero-flicker differential UI rendering engine.



Future Scope (v2.0)



WiFi NTP Sync: ESP8266 UART integration for automatic atomic time synchronization.



BLE Companion App: Android/iOS app via HC-05/BLE for remote alarm scheduling.



OLED / TFT Display: Upgrade to full-color SPI ST7789 display.



Team 2 & NTI Credits

Thank You!

Under the Supervision of

Eng. Mahmoud Morsy

[Open for Technical Questions & Answers](#)